

MODULE - II

Series Solution & Special Functions

Series Solution of Second Order Differential Equations

The solution of ordinary linear differential equation of second order with variable coefficients in the form of an infinite convergent series is called Series solution of second order differential equation.

For Example:-

Let the second order differential equation be,

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$$

$$\Rightarrow (D^2 - 2D + 1) y = 0$$

$$\Rightarrow (D - 1)^2 = 0$$

$$\Rightarrow D = 1, 1$$

$$y = (C_1 + xC_2)e^x$$

The expansion is,

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots$$

$$xe^x = x + \frac{x^2}{1!} + \frac{x^3}{2!} + \dots$$

$$\therefore y = C_1 \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \right) + C_2 \left(x + \frac{x^2}{1!} + \dots \right)$$

Ordinary Point

A point $x=a$ is called an ordinary point of the differential equation of second order,

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0$$

where, $P(x)$ & $Q(x)$ are polynomial, if both $P(x)$ & $Q(x) \rightarrow \infty$ at $x=a$.

Question: Show that $x=0$ is an ordinary point for the differential equation,

$$(1+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$$

Solⁿ: The differential eqⁿ. is,

$$(1+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} + \frac{x}{(1+x^2)} \cdot \frac{dy}{dx} - \frac{1}{(1+x^2)} y = 0$$

Comparing above eqⁿ. with standard form,
we get,

$$P(x) = \frac{x}{1+x^2} \quad \& \quad Q(x) = \frac{-1}{1+x^2}$$

$$\text{At } x=0, P(x)=0 \quad \& \quad Q(x)=-1$$

Since, both $P(x)$ & $Q(x) \rightarrow \infty$ at $x=0$.

$\therefore x=0$ is an ordinary point for the given second order differential equation.

For Ordinary Point

Q. Solve: $(1-x^2)y'' - xy' + 4y = 0$ in series.

Solⁿ:

The given eqⁿ. is,

$$(1-x^2)y'' - xy' + 4y = 0 \quad \text{--- (1)}$$

$$\Rightarrow y'' - \frac{x}{(1-x^2)} y' + \frac{4}{(1-x^2)} y = 0$$

Comparing with standard form, we get,

$$P = \frac{-x}{1-x^2} \quad \& \quad Q = \frac{4}{1-x^2}$$

$x=0$ is an ordinary point of eqⁿ. (1).

Let, $y = \sum_{k=0}^{\infty} a_k x^k$ --- (2) be the solution

$$\text{Then, } y' = \sum_{k=0}^{\infty} a_k \cdot k \cdot x^{k-1}$$

$$\& \quad y'' = \sum_{k=0}^{\infty} a_k (k)(k-1) x^{k-2}$$

Putting in eqⁿ. (1), we get,

$$\sum_{k=0}^{\infty} a_k k(k-1) x^{k-2} - \sum_{k=0}^{\infty} a_k k(k-1) x^k$$

$$-\sum_{k=0}^{\infty} a_k \cdot k \cdot x^k + \sum_{k=0}^{\infty} 4a_k x^k = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} a_k k(k-1) x^{k-2} - \sum_{k=0}^{\infty} \{k(k-1) + k - 4\} a_k x^k = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} a_k k(k-1) x^{k-2} - \sum_{k=0}^{\infty} (k^2 - 4) a_k x^k = 0$$

Putting $k=k+2$ in first term,

$$\sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) x^k - \sum_{k=0}^{\infty} (k^2 - 4) a_k x^k = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \{(k+1)(k+2) a_{k+2} - (k^2 - 4) a_k\} x^k = 0$$

$$\Rightarrow (k+1)(k+2) a_{k+2} - (k^2 - 4) a_k = 0$$

$$\Rightarrow a_{k+2} = \frac{k^2 - 4}{(k+1)(k+2)} \cdot a_k$$

For $k=0$,

$$a_2 = \frac{-4}{1 \cdot 2} \cdot a_0 = -2a_0$$

For $k=1$,

$$a_3 = \frac{-3}{2 \cdot 3} \cdot a_1 = -\frac{a_1}{2}$$

For $k=2$,

$$a_4 = \frac{0}{3 \cdot 5} \cdot a_2 = 0$$

For $k=3$,

$$a_5 = \frac{5}{4 \cdot 5} \cdot a_3 = \frac{a_3}{4} = \frac{-a_1}{8}$$

$\vdots$

Putting these values in eqⁿ. (2),

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

$$\Rightarrow y = a_0 + a_1 x - 2a_0 x^2 - \frac{a_1}{2} x^3 + 0 - \frac{a_1}{8} x^5 + \dots$$

$$\Rightarrow y = a_0(1 - 2x^2) + a_1 \left(x - \frac{x^3}{2} - \frac{x^5}{8} - \dots \right)$$

Singular Point

A point $x=a$ is said to be singular point of the second order differential equation,

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x) y = 0,$$

if at least one of $P(x)$ & $Q(x) \rightarrow \infty$ at $x=a$.

There are two types of singular point:

1. Regular Singular Point:

A singular point $x=a$ is said to be regular singular point of second order differential equation,

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x) y = 0$$

if both $(x-a)P(x)$ & $(x-a)^2 Q(x) \rightarrow \infty$ at $x=a$.

2. Irregular Singular Point:

A singular point which is not regular is known as irregular singular point.

Question: Show that $x=0$ is a regular singular point of $2x^2y'' + 3xy' + (x^2-4)y = 0$.

Solⁿ:

The given differential eqⁿ. is,

$$2x^2y'' + 3xy' + (x^2-4)y = 0$$

$$\Rightarrow y'' + \frac{3}{2x}y' + \frac{x^2-4}{2x^2}y = 0$$

Comparing with standard form, we get,

$$P(x) = \frac{3}{2x} \quad \& \quad Q(x) = \frac{x^2-4}{2x^2}$$

Now,

$$(x-a)P(x) = xP(x) = \frac{3}{2}$$

$$(x-a)^2Q(x) = x^2Q(x) = -2$$

Since, both does not tends to infinity.

$\therefore x=0$ is a regular singular point of given equation.

Frobenius Method

Case - I: When roots are different and not differing by an integer,

$$y = C_1 (y)_{m=m_1} + C_2 (y)_{m=m_2}.$$

Case - II: When roots are same then,

$$y = C_1 (y)_{m=m_1} + C_2 \left(\frac{\partial y}{\partial x} \right)_{m=m_1}$$

Case - III: When roots are distinct and differing by an integer,

$$y = C_1 (y)_{m=m_1} + C_2 \left(\frac{\partial y}{\partial x} \right)_{m=m_2}$$

Case - IV: When roots are distinct and differing by an integer, making some coefficient indeterminate, then,

$$y = C_1 (y)_{m=m_1} + C_2 (y)_{m=2}.$$

Q.1) Find the series solution of the differential equation: $3xy'' + 2y' + y = 0$.

Solⁿ:

The differential equation is,

$$3xy'' + 2y' + y = 0 \quad \text{--- (1)}$$

$$\Rightarrow y'' + \frac{2y'}{3x} + \frac{1}{3x} \cdot y = 0$$

Comparing with the standard form,

$$y'' + P(x) \cdot y' + Q(x) \cdot y = 0, \text{ we get,}$$

$$P = \frac{2}{3x} \quad \& \quad Q = \frac{1}{3x}$$

At $x=0$, $P \rightarrow \infty$ & $Q \rightarrow \infty$, so this is not an ordinary point at $x=0$.

Now,

$$xP = \frac{2}{3} \quad \& \quad x^2Q = \frac{1x}{3} = 0 \text{ at } x=0$$

So, $x=0$ is a regular singular point.

Let,

$$y = \sum_{k=0}^{\infty} a_k x^{m+k} \text{ ————— (2), be the solution.}$$

Now,

$$y' = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1} \text{ ————— (3)}$$

$$y'' = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2} \text{ ————— (4)}$$

Putting these values in eqⁿ. (1), we get,

$$3x \left[\sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2} \right]$$

$$+ 2 \left[\sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1} \right]$$

$$+ \left[\sum_{k=0}^{\infty} a_k x^{m+k} \right] = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} 3(m+k)(m+k-1) a_k x^{m+k-1}$$

$$+ \sum_{k=0}^{\infty} 2(m+k) a_k x^{m+k-1} + \sum_{k=0}^{\infty} a_k x^{m+k} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} [3(m+k)(m+k-1) + 2(m+k)] a_k x^{m+k-1} + \sum_{k=0}^{\infty} a_k x^{m+k} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} (m+k)(3m+3k-3+2) a_k x^{m+k-1} + \sum_{k=0}^{\infty} a_k x^{m+k} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} (m+k)(3m+3k-1) a_k x^{m+k-1} + \sum_{k=0}^{\infty} a_k x^{m+k} = 0 \quad (5)$$

Indicial Equation

Now, the lowest degree term in eqⁿ. (5) is obtained by putting $k=0$ in first summation.

Then, equating the coefficient of lowest power of x i.e. x^{m-1} to zero, we get,

$$m(3m-1)a_0 = 0$$

$$\Rightarrow m=0, m=\frac{1}{3} \quad [a_0 \neq 0]$$

i.e. $m=0$ & $m=1/3$ are the roots of the Indicial Equation.

Here, the roots are distinct and not differ by an integer i.e. $0 - 1/3 \neq \text{Integer}$.
indicial eqn.

Putting, $k = k+1$, in eqⁿ (5) and equating the coefficient of x^{m+k} to zero, we get,

$$(m+k+1)(3m+3k+2)a_{k+1} + a_k = 0$$

$$\Rightarrow a_{k+1} = \frac{-a_k}{(m+k+1)(3m+3k+2)}$$

For $k=0$,

$$a_1 = \frac{-a_0}{(m+1)(3m+2)}$$

For $k=1$,

$$a_2 = \frac{-a_1}{(m+2)(3m+5)} = \frac{a_0}{(m+1)(m+2)(3m+2)(3m+5)}$$

For $k=2$,

$$a_3 = \frac{-a_2}{(m+3)(3m+8)}$$

$$\Rightarrow a_3 = \frac{-a_0}{(m+1)(m+2)(m+3)(3m+2)(3m+5)(3m+8)}$$

For $m=0$,

$$a_1 = -\frac{1}{2} a_0, \quad a_2 = \frac{1}{20} a_0, \quad a_3 = -\frac{1}{480} a_0, \dots$$

Putting these values in eqⁿ. (2), we get,

$$y = x^m (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots)$$

$$\Rightarrow (y)_{m=0} = x^0 \left[a_0 + \left(-\frac{1}{2} a_0 \right) x + \left(\frac{1}{20} a_0 \right) x^2 + \left(-\frac{1}{480} a_0 \right) x^3 + \dots \right]$$

$$\Rightarrow (y)_{m=0} = a_0 \left[1 - \frac{1}{2} x + \frac{1}{20} x^2 - \frac{1}{480} x^3 + \dots \right]$$

For $m = 1/3$,

$$a_1 = -\frac{1}{4} a_0, \quad a_2 = \frac{1}{56} a_0, \quad a_3 = -\frac{1}{1680} a_0, \dots$$

Putting these values in eqⁿ. (2), we get,

$$y = x^m (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots)$$

$$\Rightarrow (y)_{m=1/3} = x^{1/3} \left[a_0 + \left(-\frac{1}{4} a_0 \right) x + \left(\frac{1}{56} a_0 \right) x^2 + \left(-\frac{1}{1680} a_0 \right) x^3 + \dots \right]$$

$$\Rightarrow (y)_{m=1/3} = a^0 \left[x^{1/3} - \frac{1}{4} x^{4/3} + \frac{1}{56} x^{7/3} - \frac{1}{1680} x^{10/3} + \dots \right]$$

Hence, the complete solution is,

$$y = C_1 (y)_{m=0} + C_2 (y)_{m=1/3}$$

$$\Rightarrow y = C_1 a_0 \left[1 - \frac{1}{2} x + \frac{1}{20} x^2 - \frac{1}{480} x^3 + \dots \right] + C_2 a_0 \left[x^{1/3} - \frac{1}{4} x^{4/3} + \frac{1}{56} x^{7/3} - \dots \right]$$

$$\Rightarrow y = A \left[1 - \frac{1}{2} x + \frac{1}{20} x^2 - \frac{1}{480} x^3 + \dots \right]$$

$$+ B \left[x^{1/3} - \frac{1}{4} x^{4/3} + \frac{1}{56} x^{7/3} - \frac{1}{1680} x^{10/3} + \dots \right]$$

where, $A = C_1 a_0$ & $B = C_2 a_0$.

Q.2) Solve the differential equation:

$$2x^2 \frac{dy^2}{dx^2} + x \frac{dy}{dx} - (x+1)y = 0$$

in power series about $x=0$.

Solⁿ:

The given differential equation is,

$$2x^2 \frac{dy^2}{dx^2} + x \frac{dy}{dx} - (x+1)y = 0 \quad \text{--- (1)}$$

$$\Rightarrow \frac{dy^2}{dx^2} + \frac{1}{2x} \cdot \frac{dy}{dx} - \frac{(x+1)}{2x^2} y = 0$$

Comparing above eqⁿ. with, $\frac{dy^2}{dx^2} + P \frac{dy}{dx} + Qy = 0$,
we get,

$$P = \frac{1}{2x} \quad \& \quad Q = -\frac{(x+1)}{2x^2}$$

At $x=0$, $P \rightarrow \infty$ & $Q \rightarrow \infty$, ~~for~~ the
eqⁿ. (1) ~~is~~ ~~not~~ $x=0$ is not an ordinary
point.

Now,

$$xP = \frac{1}{2} \quad \& \quad x^2Q = -\frac{(x+1)}{2}$$

So, $x=0$ is a regular singular point.

Let, $y = \sum_{k=0}^{\infty} a_k x^{m+k}$ — (2) be a solⁿ.

i.e. $y = x^m [a_0 + a_1 x' + a_2 x^2 + \dots]$

Then,

$$y' = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1}$$

$$\& y'' = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2}$$

Putting these values in eqⁿ. (1), we get,

$$2x^2 \left[\sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2} \right]$$

$$+ x \left[\sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1} \right] - (x+1) \left[\sum_{k=0}^{\infty} a_k x^{m+k} \right]$$

$$= 0$$

$$\Rightarrow 2 \left[\sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k} \right]$$

$$+ \left[\sum_{k=0}^{\infty} a_k (m+k) x^{m+k} \right] - \left[\sum_{k=0}^{\infty} a_k x^{m+k+1} \right] - \left[\sum_{k=0}^{\infty} a_k x^{m+k} \right]$$

$$= 0$$

$$\Rightarrow \sum_{k=0}^{\infty} [2(m+k)(m+k-1) + (m+k) - 1] a_k x^{m+k}$$

$$- \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

~~$$\Rightarrow \sum_{k=0}^{\infty} [(m+k)[2m+2k-2+1] - 1] a_k x^{m+k}$$~~

~~$$- \sum_{k=0}^{\infty} a_k x^{m+k+1} \neq 0$$~~

$$\Rightarrow \sum_{k=0}^{\infty} [2(m+k)(m+k) - 2(m+k) + (m+k) - 1] a_k x^{m+k}$$

$$- \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} [2(m+k)^2 - 2m - 2k + m + k - 1] a_k x^{m+k}$$

$$- \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} [2(m+k)^2 - (m+k) - 1] a_k x^{m+k}$$

$$- \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} [2(m+k)^2 - 2(m+k) + (m+k) - 1] a_k x^{m+k}$$

$$- \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \left\{ 2(m+k)[(m+k)-1] + 1[(m+k)-1] \right\} a_k x^{m+k} - \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \left\{ [2(m+k)+1][(m+k)-1] \right\} a_k x^{m+k} - \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} (m+k-1)(2m+2k+1) a_k x^{m+k} - \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0 \quad (3)$$

Indicial Eqⁿ.

Here, the lowest degree term in Indicial Equation is obtained by putting $k=0$.

Putting $k=0$ in eqⁿ's (5) Indicial Equation and equating the coefficient of x^m to zero, we get,

$$(m-1)(2m+1)a_0 = 0$$

$$\Rightarrow m=1, 'm=-1/2$$

Here, the roots are different and doesn't differ by an integer.

Putting, $k=k+1$ in the indicial equation and equating the coefficient of x^{m+k+1} to zero, we get,

$$a_{k+1} [(m+k)(2m+2k+1) - 1] = 0$$

$$\Rightarrow (m+k)(2m+2k+1) a_{k+1} - a_k = 0$$

$$\Rightarrow a_{k+1} = \frac{a_k}{(m+k)(2m+2k+3)}$$

Putting, $k=0, 1, 2, \dots$, we get,

$$a_1 = \frac{a_0}{m(2m+3)}$$

$$a_2 = \frac{a_1}{(m+1)(2m+5)} = \frac{a_0}{m(m+1)(2m+5)(2m+3)}$$

$$a_3 = \frac{a_2}{(m+2)(2m+7)} = \frac{a_0}{m(m+1)(m+2)(2m+7)(2m+3)(2m+5)}$$

$\vdots$

Putting, $m=1$, we get,

$$a_1 = \frac{a_0}{5}, \quad a_2 = \frac{a_0}{70}, \quad a_3 = \frac{a_0}{1890}, \dots$$

Putting these in eqⁿ. (2), we get,

$$\text{i.e. } y = x^m (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots)$$

$$\Rightarrow (y)_{m=1} = x a_0 \left[1 + \frac{1}{m(2m+3)} x \right]$$

$$\Rightarrow (y)_{m=1} = x' \left[a_0 + \frac{a_0}{5} x + \frac{a_0}{70} x^2 + \frac{a_0}{1890} x^3 + \dots \right]$$

$$\Rightarrow (y)_{m=1} = x a_0 \left[1 + \frac{1}{5} x + \frac{1}{70} x^2 + \frac{1}{1890} x^3 + \dots \right]$$

Now, putting $m = -1/2$, in $a_1, a_2, a_3 \dots$, we get,

$$a_1 = -a_0, a_2 = -\frac{a_0}{2}, a_3 = -\frac{a_0}{18}, \dots$$

Putting these values in eqⁿ. (2), we get,

$$(y)_{m=-1/2} = x^{-1/2} \left[+a_0 - \frac{a_0}{2} x - \frac{a_0}{18} x^2 - \dots \right]$$

$$\Rightarrow (y)_{m=-1/2} = -a_0 \left[-x^{-1/2} + x^{1/2} + \frac{1}{20} x^{3/2} + \dots \right]$$

Hence, the complete solution is,

$$y = C_1 (y)_{m=1} + C_2 (y)_{m=-1/2}$$

$$\Rightarrow y = C_1 x a_0 \left[1 + \frac{1}{5} x + \frac{1}{70} x^2 + \frac{1}{1890} x^3 + \dots \right]$$

$$+ C_2 (-a_0) \left[-x^{-1/2} + \cancel{x^{1/2}} + \frac{1}{20} x^{3/2} + \dots \right]$$

$$\Rightarrow y = A \left[x + \frac{1}{5} x^2 + \frac{1}{70} x^3 + \frac{1}{1890} x^4 + \dots \right]$$

$$- B \left[-x^{-1/2} + \cancel{x^{1/2}} + \frac{1}{20} x^{3/2} + \dots \right]$$

where, $A = C_1 a_0$ & $B = C_2 a_0$.

Q3) Solve: $x^2 y'' + 5xy' + x^2 y = 0$ in series.

Solⁿ:

The given differential equation is,

$$x^2 y'' + 5xy' + x^2 y = 0 \quad \text{--- (1)}$$

$$\Rightarrow y'' + \frac{5}{x} y' + y = 0$$

Comparing with standard form, $P = \frac{5}{x}$ & $Q = 1$

~~At $x=0$~~ , $xP = 5$ & $x^2 Q = x^2$

So, $x=0$ is a regular singular point of the eqⁿ. (1).

Let, $y = \sum_{k=0}^{\infty} a_k x^{m+k}$ --- (2), be a solution.

$$\text{i.e. } y = x^m (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots) \quad \text{--- (3)}$$

Then,

$$y' = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1}$$

$$y'' = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2}$$

Putting these in eqⁿ. (1), we get,

$$\sum_{k=0}^{\infty} a_k(m+k)(m+k-1)x^{m+k} + \sum_{k=0}^{\infty} 5(m+k)a_k x^{m+k} + \sum_{k=0}^{\infty} a_k x^{m+k+2} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} [(m+k)(m+k-1) + 5(m+k)] a_k x^{m+k} + \sum_{k=0}^{\infty} a_k x^{m+k+2} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} (m+k)[(m+k-1) + 5] a_k x^{m+k} + \sum_{k=0}^{\infty} a_k x^{m+k+2} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} (m+k)(m+k+4)a_k x^{m+k} + \sum_{k=0}^{\infty} a_k x^{m+k+2} = 0 \quad (4)$$

Now, the Indicial Equation is,

$$(m+0)(m+0+4)a_0 = 0$$

$$\Rightarrow m(m+4)a_0 = 0$$

$$\Rightarrow m = 0, -4, a_0 \neq 0$$

Here, the roots are different and differ by an integer i.e. -4 .

Now, if $k=1$,

$$(m+1)(m+1+4)a_1 = 0$$

$$\Rightarrow a_1 = 0$$

if $k = k+2$, we get,

$$(m+k+2)(m+k+6)a_{k+2} + a_k = 0$$

$$\Rightarrow a_{k+2} = \frac{-a_k}{(m+k+2)(m+k+6)}$$

if $k=0, 1, 2, 3, \dots$, we get,

$$a_2 = \frac{-a_0}{(m+2)(m+6)}$$

$$a_3 = \frac{-a_1}{(m+3)(m+7)} = 0$$

$$a_5 = a_7 = 0$$

$$a_4 = \frac{-a_2}{(m+4)(m+8)} = \frac{a_0}{(m+2)(m+4)(m+6)(m+8)}$$

We have,

$$a_1 = a_3 = a_5 = a_7 = \dots = 0$$

$$a_2 = \frac{-a_0}{(m+2)(m+6)}, \quad a_4 = \frac{a_0}{(m+2)(m+4)(m+6)(m+8)}, \dots$$

Putting these values in eqⁿ. (3), we get,

$$(y)' = x^3 \left[a_0 + 0 + \left(\frac{-a_0}{(m+2)(m+6)} \right) x^2 + 0 + \left(\frac{a_0}{(m+2)(m+4)(m+6)(m+8)} \right) x^4 + \dots \right] \quad (5)$$

For $m=0$,

$$\Rightarrow (y)'_{m=0} = a_0 \left[1 - \frac{1}{2 \cdot 6} x^2 + \frac{1}{2 \cdot 4 \cdot 6 \cdot 8} x^4 - \dots \right]$$

$$\Rightarrow (y)'_{m=0} = a_0 \left[1 - \frac{x^2}{12} + \frac{x^4}{384} - \dots \right] \quad (5)$$

For $m=-4$, the coefficients becomes infinite i.e. the coefficient of x^4 and so on. ~~and so on.~~

To remove this we multiply put $b_0(m+4)=a_0$, so that the term making coefficients infinite cancel out.

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$$y = \underbrace{b_0 x^m}_u \left[(m+4) - \frac{(m+4)}{(m+2)(m+6)} x^2 \right]$$

$$+ \frac{1}{(m+2)(m+6)(m+8)} x^4 - \dots \Big]$$

$$\Rightarrow \frac{dy}{dm} = b_0 x^m \log x \left[(m+4) - \frac{(m+4)x^2}{(m+2)(m+6)} + \frac{x^4}{(m+2)(m+6)(m+8)} - \dots \right]$$

$$+ b_0 x^m \left[1 + \frac{(m^2+8m+20)}{(m^2+8m+12)} x^2 - \frac{(3m^2+32m+76)x^4}{(m^3+16m^2+76m+96)^2} - \dots \right]$$

$$\left[\frac{d(uv)}{dx} = u'v + uv' \quad \& \quad \frac{d(u/v)}{dx} = \frac{u'v - uv'}{v^2} \right]$$

$$\Rightarrow \left(\frac{dy}{dm} \right)_{m=-4} = b_0 x^{-4} \log x \left[\frac{-x^4}{16} - \dots \right]$$

$$+ b_0 x^{-4} \left[1 + \frac{x^2}{4} - \frac{x^4}{4^3} + \dots \right] \quad (7)$$

Hence, the complete solution is,

$$y = C_1 (y)_{m=0} + C_2 \left(\frac{dy}{dm} \right)_{m=-4} \quad \underline{\text{Ans}}$$

Q4) Solve in series, the differential equation:

$$xy'' + 2y' + xy = 0$$

Solⁿ:

The given eqⁿ is,

$$xy'' + 2y' + xy = 0 \quad \text{--- (1)}$$

$$\Rightarrow y'' + \frac{2}{x}y' + xy = 0$$

Comparing with standard form, we get,
 $P = \frac{2}{x}$ & $Q = x$

$x=0$ is a regular singular point because,
 $xP = 2$ & $x^2Q = x^3$ does not tends to infinity.

Let, $y = \sum_{k=0}^{\infty} a_k x^{m+k}$ be a solution.

$$\text{i.e. } y = x^m (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots) \quad (2)$$

$$\text{Now, } y' = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1}$$

$$\& \quad y'' = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2}$$

Putting these values in eqⁿ. (1), we get,

$$\sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-1} + \sum_{k=0}^{\infty} 2a_k (m+k) x^{m+k-1} + \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \{(m+k)(m+k-1) + 2(m+k)\} a_k x^{m+k-1} + \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} (m+k)(m+k+1) a_k x^{m+k-1} + \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0 \quad (3)$$

The coefficient of x^{m+0-1} i.e. x^{m-1} is,

$$(m+0)(m+0+1)a_0 = 0$$

$$\Rightarrow m(m+1)a_0 = 0$$

$$\Rightarrow m = 0, -1 \quad a_0 \neq 0$$

Here, the roots of indicial eqⁿ. is different and differ by integers.

For $k=1$,

$$(m+1)(m+2)a_1 = 0$$

$$\Rightarrow a_1 = \frac{0}{0} \quad \text{for } m = -1$$

$\Rightarrow a_1$ becomes arbitrary at $m = -1$.

For $k = k+2$,
 $(m+k+2)(m+k+3)a_{k+2} + a_k = 0$

$$\Rightarrow a_{k+2} = \frac{-a_k}{(m+k+2)(m+k+3)}$$

for $k=0, 1, 2, 3, \dots$,

$$a_2 = \frac{-a_0}{(m+2)(m+3)}$$

$$a_3 = \frac{-a_1}{(m+3)(m+4)}$$

$$a_4 = \frac{-a_2}{(m+4)(m+5)} = \frac{a_0}{(m+2)(m+3)(m+4)(m+5)}$$

$$a_5 = \frac{-a_3}{(m+5)(m+6)} = \frac{a_1}{(m+3)(m+4)(m+5)(m+6)}$$

$\vdots$

Putting in eqⁿ. (2), we get,

$$y = x^m \left[a_0 \left(1 - \frac{x^2}{(m+2)(m+3)} + \frac{x^4}{(m+2)(m+3)(m+4)(m+5)} - \dots \right) \right. \\ \left. + a_1 \left(x - \frac{x^3}{(m+3)(m+4)} + \frac{x^5}{(m+3)(m+4)(m+5)(m+6)} - \dots \right) \right]$$

$$\Rightarrow (y)_{m=-1} = x^{-1} \left[a_0 \left(1 - \frac{x^2}{1 \cdot 2} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4} - \dots \right) + a_1 \left(x - \frac{x^3}{2 \cdot 3} + \frac{x^5}{2 \cdot 3 \cdot 4 \cdot 5} - \dots \right) \right]$$

$$\Rightarrow (y)_{m=-1} = \frac{1}{x} [a_0 \cos x + a_1 \sin x]$$

Since, at $m=-1$, the solution contains two arbitrary constants. So, $y = (y)_{m=-1}$ can be complete solution.

Hence, the complete solution is,

$$y = \frac{1}{x} (a_0 \cos x + a_1 \sin x).$$

Q. Solve: $x(1-x)y'' - 3xy' - y = 0$ in series.

Solⁿ:

The given eqⁿ. is,

$$x(1-x)y'' - 3xy' - y = 0 \quad \text{--- (1)}$$

$$\Rightarrow y'' - \frac{3}{(1-x)} y' - \frac{1}{x(1-x)} y = 0$$

Comparing with standard form, we get,
 $P = \frac{-3}{(1-x)}$ & $Q = \frac{-1}{x(1-x)}$

$x=0$ is an regular singular point of eqⁿ. (1).
Because xP & x^2Q doesn't tends to infinity.

Let, $y = \sum_{k=0}^{\infty} a_k x^{m+k}$ be a solution.

$$\text{i.e. } y = x^m (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots) \quad \text{--- (2)}$$

Then,

$$y' = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1}$$

$$\& \quad y'' = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2}$$

Putting in eqⁿ. (1), we get,

$$\sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-1}$$

$$\blacklozenge - \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k} - \sum_{k=0}^{\infty} 3a_k (m+k) x^{m+k} - \sum_{k=0}^{\infty} a_k \cancel{(m+k)} x^{m+k} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-1}$$

$$- \sum_{k=0}^{\infty} \left[(m+k)(m+k-1) + 3(m+k) + \cancel{(m+k)} \right] a_k x^{m+k}$$

$$= 0$$

Putting $k=0$ in first term, $(m)(m-1)a_0 = 0 \Rightarrow m=0, 1$

→ [Putting $k=k+1$ in first summation]

$$\sum_{k=0}^{\infty} (m+k+1)(m+k) a_{k+1} x^{m+k}$$

$$- \sum_{k=0}^{\infty} \left[(m+k)(m+k-1) + 3(m+k) + 1 \right] a_k x^{m+k} = 0$$

Equating the coefficient of x^{m+k} to zero,

$$(m+k+1)(m+k) a_{k+1} - \left[(m+k)(m+k-1) + 3(m+k) + 1 \right] a_k = 0$$

$$\Rightarrow a_{k+1} = \frac{(m+k)(m+k-1) + 3(m+k) + 1}{(m+k+1)(m+k)} \cdot a_k$$

$$\Rightarrow a_{k+1} = \frac{\{(m+k)(m+k-1+3)\}+1}{(m+k+1)(m+k)} \cdot a_k$$

$$\Rightarrow a_{k+1} = \frac{(m+k)(m+k+2)+1}{(m+k+1)(m+k)} \cdot a_k$$

$$\Rightarrow a_{k+1} = \frac{(m+k)^2 + 2(m+k) + 1}{(m+k+1)(m+k)} \cdot a_k$$

$$\Rightarrow a_{k+1} = \frac{(m+k+1)^2}{(m+k+1)(m+k)} \cdot a_k$$

$$\Rightarrow a_{k+1} = \frac{m+k+1}{m+k} \cdot a_k$$

For $k=0, 1, 2, 3, \dots$,

$$a_{0+1} = a_1 = \frac{m+1}{m} \cdot a_0$$

$$a_{1+1} = a_2 = \frac{m+2}{m+1} \cdot a_1 = \frac{(m+1)(m+2)}{m(m+1)} \cdot a_0$$

$$a_{2+1} = a_3 = \frac{m+3}{m+2} \cdot a_2 = \frac{(m+1)(m+2)(m+3)}{m(m+1)(m+2)} \cdot a_0$$

⋮

Putting these values in eqⁿ. (2),

$$y = x^m \left[a_0 + \frac{(m+1)}{m} a_0 x + \frac{(m+1)(m+2)}{m(m+1)} a_0 x^2 + \dots \right] \quad (3)$$

$$\Rightarrow (y)_{m=1} = x a_0 (1 + 2x + 3x^2 + \dots) \quad (4)$$

If we put $m=0$ then the coefficients become indeterminate. So, we let $b_0 = a_0 m$.

$$(y) = b_0 x^m \left[m + (m+1)x + \frac{(m+1)(m+2)}{(m+1)} x^2 + \dots \right]$$

$$\Rightarrow y = \underbrace{b_0 x^m}_u \left[\underbrace{m + (m+1)x + (m+2)x^2 + \dots}_v \right]$$

$$\Rightarrow \frac{dy}{dm} = b_0 x^m \log x [m + (m+1)x + (m+2)x^2 + \dots] + b_0 x^m [1 + x + x^2 + \dots]$$

$$\Rightarrow \left(\frac{dy}{dm} \right)_{m=0} = b_0 \log x (x + 2x^2 + 3x^3 + \dots) + b_0 (1 + x + x^2 + \dots) \quad (5)$$

Hence, the complete solution is,

$$y = C_1 (y)_{m=1} + C_2 \left(\frac{dy}{dm} \right)_{m=0}$$

$$\Rightarrow y = a_0 x (1 + 2x + 3x^2 + \dots) + b_0 \log x (x + 2x^2 + \dots) + b_0 (1 + x + x^2 + \dots)$$